INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/017305

A. CLASSIFICATION OF SUBJECT MATTER
INV. G02B1/14 A63J5/00 C03C17/02 C23C14/10 G02B5/08
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G02B A63D C23C C03C A63J

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US 2020/096682 A1 (TESHIMA MASAHIRO [DE] ET AL) 26 March 2020 (2020-03-26)	1,3-5
A	paragraph [0027] - paragraph [0031]; figures 1a, 4f, 5f	2
Y	US 5 619 375 A (ROBERTS JOHN K [US]) 8 April 1997 (1997-04-08) column 8 - column 9; figure 3	1,3-5
A	US 2013/208375 A1 (KRASNOV ALEXEY [US] ET AL) 15 August 2013 (2013-08-15) paragraph [0028]	4
A	US 6 065 840 A (CASKEY GREGORY T [US] ET AL) 23 May 2000 (2000-05-23) figures 9a, 10, 11	2

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
5 July 2022	08/09/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Denner, Peter

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US2022/017305

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:
1-5

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-5

Dual-surfaced mirror system

---

2. claims: 6-20

Mirror wedge system

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2022/017305

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2020096682 A1	26-03-2020	CN 110506223 A	26-11-2019
		EP 3339918 A1	27-06-2018
		EP 3559711 A1	30-10-2019
		JP 6927536 B2	01-09-2021
		JP 2020502582 A	23-01-2020
		US 2020096682 A1	26-03-2020
		WO 2018115276 A1	28-06-2018
US 5619375 A	08-04-1997	EP 0855603 A1	29-07-1998
		US 5528422 A	18-06-1996
		US 5619374 A	08-04-1997
		US 5619375 A	08-04-1997
US 2013208375 A1	15-08-2013	NONE	
US 6065840 A	23-05-2000	US 6065840 A	23-05-2000
		US 6196688 B1	06-03-2001